Neural Architecture Search on NATS Bench

AutoML – Assignment 2

Leiden University

1 Tutorial

Familiarise yourself with the NATS-Bench paper and Python package [2]. Install `nats_bench` in a Python environment, and download the tutorial that is available on BrightSpace. Work through the various components, such as:

- Setting up the API (fast mode loads faster and specifically for laptops, but the other mode works better if you have the RAM available on your machine)
- Acquiring the performance data of a specific architecture
- Plotting the DAG of this architecture
- Performing random search over architectures
- Plotting the learning curve of specific architectures

Note that various versions have different keys. It is probably easiest to have a debugger available. Note that you do not need to work in the Jupyter notebook, you can also work in your favourite IDE. However, Jupyter might be useful, because the kernel can leave the loaded API in memory.

During the Monday sessions (2x) you are entitled to support from our team on these operations.

2 Assignment

Implement the following Neural Architecture Search methods using this framework:

- Random Search
- Random Sample
- Bayesian Optimisation (put some thoughts how to represent information for the surrogate model and how to define an acquition function)
- Successive Halving
- An iteration-based learning curve method (see, e.g., the Power-law version of [1]).

Compare the performance of these methods using some reasonable standards (e.g., give them the same budget) on various pre-computed datasets.

Note that while the runtime of these experiments would have been extensive,

3 Hand-in

Hand in the following material:

- An easy to understand code-base, making efficient use of the various functions that are available in NATS Bench; abstrating away from various complexities
- A README explaining the built-up of the Code Bench.
- A scientific rapport (max 4 pages excluding references), using the provided LaTeX template, in which you follow all elements that are laid out in the guidelines.
- In order to comply with the page limit, you can't go into too much detail explaining each individual method. Provide a brief description per method plus reference to a paper.

References

[1] Tobias Domhan, Jost Tobias Springenberg, and Frank Hutter. Speeding up automatic hyperparameter optimization of deep neural networks by extrapolation of learning curves. In Qiang Yang and Michael J. Wooldridge, editors, *Proceedings of the Twenty-Fourth International Joint Conference on Artificial Intelligence, IJCAI 2015, Buenos Aires, Argentina, July 25-31, 2015*, pages 3460–3468. AAAI Press, 2015.

[2] Xuanyi Dong, Lu Liu, Katarzyna Musial, and Bogdan Gabrys. Nats-bench: Benchmarking NAS algorithms for architecture topology and size. *IEEE Trans. Pattern Anal. Mach. Intell.*, 44(7):3634–3646, 2022.